

**TEAM NO.: 7**

**STUDENTS DETAILS:**

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**PROJECT TITLE:** Big Data Processing and Comparative Analytics for Large-Scale Medical Knowledge Discovery in the Huatuo-26M Dataset

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| <b>DOMAIN</b>          | Selected Domain                | Healthcare                                                                                                                                                                                                                                                                                                                                                  |
|                        | Description                    | This project involves the distributed processing of the Huatuo-26M dataset, the largest medical QA collection containing 26 million question-answer pairs. The study focuses on mining medical insights, disease patterns, and consultation trends using Big Data frameworks.                                                                               |
|                        | Justification                  | The dataset contains 26 million records, making it a "High-Volume" big data source that is impossible to process efficiently on traditional single-node systems. Distributed frameworks like Hadoop and Spark are required to manage the computational intensity of processing over 4.5 GB of semi-structured text data for meaningful knowledge discovery. |
| <b>PROBLEM/ ISSUES</b> | For Data Processing/ Analytics | 1. Volume and Scalability: Analysing 26 million QA pairs for disease trends requires massive computational power and scalable MapReduce logic that exceeds traditional database capabilities.                                                                                                                                                               |
|                        |                                | 2. Data Variety and Heterogeneity: Medical QA data is semi-structured and colloquial; extracting structured medical insights requires complex text-mining algorithms (e.g., Pattern Matching, Wordcount) across distributed nodes.                                                                                                                          |
|                        |                                | 3. Framework Efficiency: There is a critical need to evaluate and compare the processing time performance of Hadoop vs. Spark when handling high-velocity medical consultation records to identify the most suitable framework for healthcare AI.                                                                                                           |
| <b>DATASET</b>         | Name                           | Huatuo-26M (Medical Consultation QA Subset)                                                                                                                                                                                                                                                                                                                 |
|                        | Description                    | An integrated medical dataset containing expert doctor-patient consultation records, encyclopedia entries, and medical knowledge bases.                                                                                                                                                                                                                     |
|                        | URL/ link                      | <a href="#">Huatuo-26M (Medical Consultation QA Subset)</a>                                                                                                                                                                                                                                                                                                 |
|                        | Justification                  | The Huatuo-26M (specifically the consultation subset) is uniquely qualified for this research as it represents the largest specialized medical QA collection to date, offering a high-volume repository of 26 million expert-patient interactions. Choosing this dataset is technically                                                                     |

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|                                 |                                                  | significant because its substantial scale of 4.54 GB provides an authentic environment to evaluate the scalability and efficiency of distributed MapReduce algorithms.                                                                                                                                                                                                                                                                                                                                     |
|                                 | Size                                             | 4.54 GB (Total Consultation subset)                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                 | Others<br>-attributes<br>-no of columns<br>-etc. | Attributes: ID, Question_Sequence, Answer_Sequence, Expert_Score, Department_Label, Related_Diseases, Hospital_Org, Doctor_Name, Timestamp<br><br>No. of Columns: 25–30 columns                                                                                                                                                                                                                                                                                                                            |
|                                 |                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>CASE STUDIES DESCRIPTION</b> | Case Study 1                                     | <b>Title:</b> Medical Department Demand and Resource Allocation Analysis<br><b>Description:</b> Analysing which medical departments (e.g., Pediatrics, Oncology) have the highest volume of patient queries to identify critical healthcare service demands.<br><b>Technical Implementation:</b> We will implement Filtering (to isolate specific departments), Aggregation/Count (to sum query volumes), Sorting (to rank demand levels), and Retrieval (to extract top-priority medical cases).          |
|                                 | Case Study 2                                     | <b>Title:</b> Disease Prevalence and Correlation Pattern Matching.<br><b>Description:</b> Identifying high-frequency disease mentions and their correlations within the 26 million records to map medical trends.<br><b>Technical Implementation:</b> We will implement WordCount (to find disease keywords), Pattern Matching (to find symptom associations), Joining (to link questions with answer labels), and Grouping (by disease category).                                                         |
|                                 | Case Study 3                                     | <b>Title:</b> Expert Advice Quality and Sentiment Benchmarking.<br><b>Description:</b> Evaluating the "Expert Scores" across millions of records to determine the consistency and quality of medical advice provided in different hospitals.<br><b>Technical Implementation:</b> We will implement Average/Mean Calculation (for quality scores), Partitioning (to distribute processing by Hospital Org), Min/Max Retrieval (to find highest-rated advice), and Filtering (to remove low-score outliers). |
|                                 | Case Study 4                                     | <b>Title:</b> Cross-Regional Healthcare Pattern Analytics<br><b>Description:</b> Comparing medical consultation patterns across different hospital organizations to identify regional health concerns.<br><b>Technical Implementation:</b> We will implement Grouping (by Hospital Organization), Joining (to combine metadata with QA text), Retrieval (to fetch                                                                                                                                          |

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|  |  | regional disease trends), and Calculation (of department-to-patient ratios). |
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